THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

May 8, 2017 14:30 pm – 17:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. a) Management contracting is regarded as a procurement method which can 'fast track' a construction project. With the aid of suitable figure(s), describe the concept of management contracting and discuss whether the government should consider using this kind of procurement method to speed up the public housing programme in Hong Kong. [13 marks]

   b) Explain with detailed justifications other suitable procurement methods which can satisfy the huge demand for public housing in Hong Kong within a short period of time. [12 marks]

2. A 2-kilometer road will be constructed in sections of 25 meters long in stages. While the total construction period of this road project is 500 working days, the work sequence and time for constructing one typical road section are as shown in the following table:

Activity	*Duration (working day)*
A	5
B	5
C	10
D	5
E	5

   a) Based on the above information, calculate how long should be allowed for the 'turnover' of activities. [3 marks]

   b) Prepare a suitable programme based on the data and findings in sub-sections (a). [10 marks]

   c) Explain why planners should calculate the 'turnover' and prepare the programme as required in sub-sections (a) and (b) respectively. [5 marks]

   d) Describe the measures to be introduced by planners to enhance project coordination in order to ensure the 'turnover' as identified in sub-section (a) is strictly adhered to. [7 marks]

3. a) Explain with relevant examples why civil engineering projects are susceptible to the risks of delay and cost overrun. [10 marks]

   b) Describe the roles of civil engineers in different project stages and discuss how civil engineers can help ensure the project objectives are satisfactorily achieved. [15 marks]

4. An investor is interested to invest in either Scheme 'A' or Scheme 'B', and the projected cash flow (all values in $ million) of the two schemes as follows:

Scheme 'A'
Investment period: 8 years
Capital cost: $100
Spending: $15 in year 5; $5 in year 7
Income in years 1-8: $30, $25, $20, $15, $20, $18, $16 and $15
Reversion value: $35

Scheme 'B'
Investment period: 10 years
Capital cost: $100
Spending: $8 in year 6
Income in years 1-10: $20, $22, $24, $22, $20, $22, $20, $18, $16 & $16
Reversion value: $0

a) Using a discount rate of 8%, calculate the net present value of both Schemes 'A' and 'B'. [8 marks]

b) Compute the internal rate of return of both Schemes 'A' and 'B', and determine which is a more attractive scheme based on the internal rate of return. [10 marks]

c) Briefly explain the differences between the net present value and internal rate of return methods, and discuss which method should the investor employ to facilitate him/her arriving at a sensible investment decision. [7 marks]

5. Calculate the unit rate of a 10,000 × 3,000 × 100 mm thick block wall in cement mortar (1:3) based on the following data:

Material
Size of block = 290 × 140 × 100 mm
Price of block = $4,000.00 per 1,000 no.
Price of cement = $1,000.00 per tonne
Price of sand = $250.00 per tonne

Labour cost
Bricklayer = $100.00 per hour
Labourer = $75.00 per hour
Unloading = 2 hours per 1,000 blocks

Labour output
Wall erection = 0.85 hours / m^2 [25 marks]

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